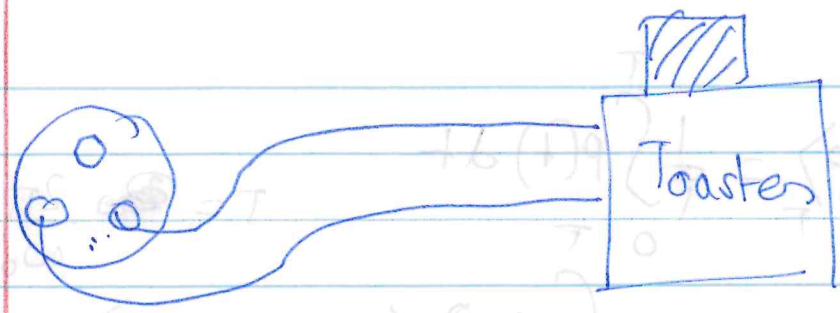
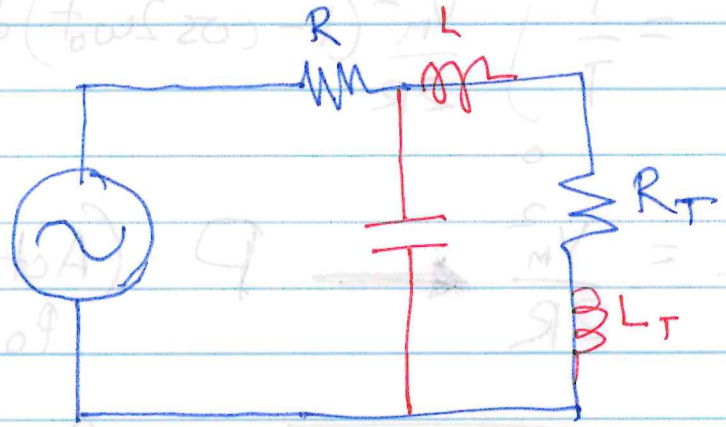
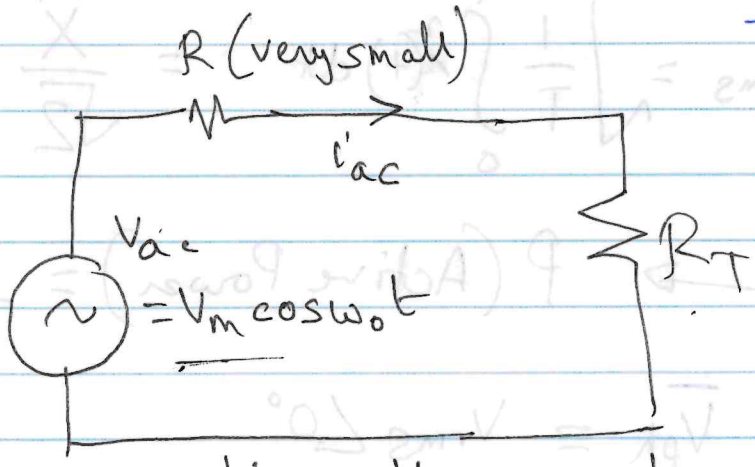




* Voltage Source
* Resistor
* Switch.
* Inductor
* Capacitor.
* Transistor



Power.?
Current.

$$V_{ac} = V_m \cos \omega_0 t$$

$$i'_{ac} = \frac{V_m}{R} \cos \omega_0 t$$

$$p(t) = V_{ac} \cdot i'_{ac} = \frac{V_m^2}{R} \cos^2 \omega_0 t$$

$$= \frac{V_m^2}{2R} (1 - \cos 2\omega_0 t)$$

$$\langle P \rangle_T = \frac{1}{T}\int_0^T p(t)\,dt \qquad T = \frac{2\pi}{\omega_0}$$

$$= \frac{1}{T}\int_0^T \frac{V_m^2}{2R}(1-\cos 2\omega_0 t)\,dt$$

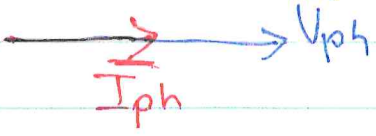
$$= \frac{V_m^2}{2R} \longleftarrow P \text{ (Active Power)}$$

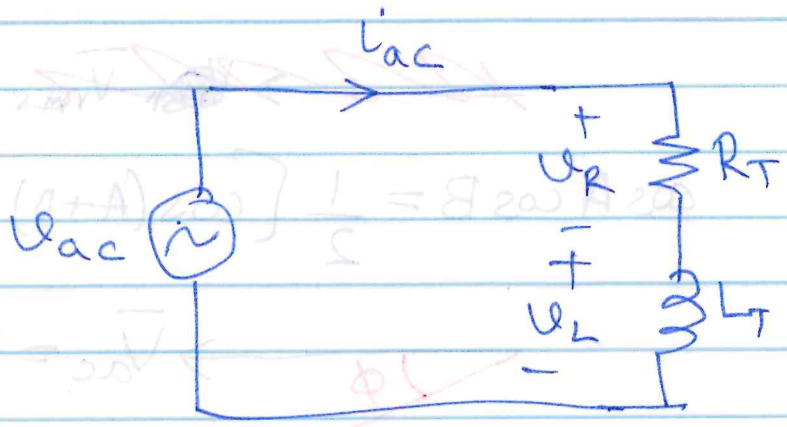
$$x_{rms} = \sqrt{\frac{1}{T}\int_0^T x^2(t)\,dt} = \frac{\hat{X}}{\sqrt{2}}$$

$$\Rightarrow P\,(\text{Active Power}) = \frac{V_{rms}^2}{R}$$

$$\bar{V}_{ph} = V_{rms}\angle 0^\circ$$

$$\bar{I}_{ph} = I_{rms}\angle 0^\circ$$





$$v_R = R_T \cdot i_{ac}$$

$$v_L = L \frac{di}{dt}$$

$$\boxed{v_{ac} = v_R + v_L}$$

$$V_m \cos\omega_0 t = v_{ac} = R_T i_{ac} + L_T \frac{di_{ac}}{dt}$$

$$\bar{V}_m = R_T \bar{I}_{ac} + j\omega_0 L_T \cdot \bar{I}_{ac}$$

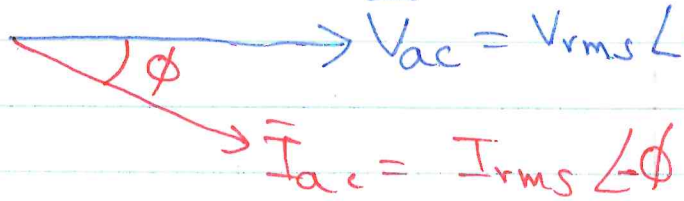
$$\bar{I}_{ac} = \frac{\bar{V}_m}{R_T + j\omega_0 L_T}$$

$$= \frac{V_{rms} \angle 0^\circ}{R_T + j\omega_0 L_T} = \frac{V_{rms}}{\sqrt{R_T^2 + \omega_0^2 L_T^2}} \angle \phi$$

$$\phi = \pm \tan^{-1}\left(\frac{\omega_0 L}{R}\right)$$

$\bar{V}_{rms}$

$$\cos A \cos B = \frac{1}{2}\left[\cos(A+B) + \cos(A-B)\right]$$



$$v_{ac} = V_m \cos \omega_0 t$$

$$i_{ac} = \frac{V_m}{\sqrt{R_T^2 + \omega_0^2 L_T^2}} \cdot \cos(\omega_0 t - \phi)$$

$$P_{ac}(t) = v_{ac}(t) \cdot i_{ac}(t)$$

$$= \frac{V_m^2}{\sqrt{R_T^2 + \omega_0^2 L_T^2}} \cos \omega_0 t \cdot \cos(\omega_0 t - \phi)$$

$$P = \frac{1}{T}\int_0^T P_{ac}(t)\, dt =$$

$$= \frac{1}{2T}\int_0^T \left\{\left[\cos\phi\right] + \left[\cos(2\omega_0 t - \phi)\right]\right\} dt$$

$$P = \frac{V_m^2}{2\sqrt{R_T^2 + \omega_0^2 L_T^2}} \cos\phi$$

$$= V_{rms} \cdot I_{rms} \cos\phi \,.$$